Logic, Language, and Philosophy

Natural deduction

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Overview

Recap natural deduction

Rules for predicate logic

Homework

Background

Recap natural deduction

Proofs

A proof of $\phi_1 \ldots \phi_n \vdash \psi$ is a finite sequence of formulas that:

- starts with premisses $\phi_1 \ldots \phi_n$
- ends with conclusion ψ
- and in which each step is motivated by appeal to one rule

Notation:

- $\phi_1 \ldots \phi_n \vdash \psi$

Rules

Two main components:

- Introduction rules: how do I *get* something of a certain form?
- Elimination ('use') rules: how can I *use* something of a certain form?

Rules for predicate logic

Preliminary

Note:

- Rules are stated in terms of *sentences*
- This is not a restriction, given:

Fact

$\models \phi$ *iff* $\models \forall x_1 \dots x_n \, \phi$, *where* x_i *free in* ϕ

- and:

Theorem (Deduction theorem)

$\phi_1 \dots \phi_n \models \psi$ *iff* $\models (\phi_1 \wedge \dots \wedge \phi_n) \rightarrow \psi$

Consequence: the substitution method and the assignment method characterise the same set of valid arguments.

Preliminary (cont.)

Recall notation:

- $[c/x]\phi$: result of substituting c for all *free* occurrences of x in ϕ
  - $[c/x]Rxx = Rcc$
  - $[c/x]Px \wedge \exists x Qx = Pc \wedge \exists x Qx$
  - $[c/x][c'/y]Rxy = [c/x]Rxc' = Rcc'$
  - $[c/x][d/x]Rxy = [c/x]Rdy = Rdy$ (order is 'inside out')

Existential quantifier introduction

Introduction rule $I_\exists$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & [c/x]\phi & \\ \vdots & & \\ n. & \exists x\phi & I_\exists, m \end{array}$$

A.k.a. “Existential Generalisation”

Examples

Example 1:

$$Raa \vdash \exists x Rxx$$

1.	Raa	$\mathtt{A}$
2.	$\exists x Rxx$	$I_{\exists}, 1$

Note: $Raa = [a/x]Rxx$

Examples (2)

Example 2:

$$Raa \vdash \exists x \exists y Rxy$$

1.	Raa	A
2.	$\exists y Ray$	$I_\exists, 1$
3.	$\exists x \exists y Rxy$	$I_\exists, 2$

Note: $Raa = [a/y]Ray$ and $\exists y Ray = [a/x]\exists y Rxy$

Examples (3)

Example 3:

$$Bb \vdash \exists x \exists x Bx$$

1.	Bb	A
2.	$\exists x Bx$	$I_\exists, 1$
3.	$\exists x \exists x Bx$	$I_\exists, 2$

Note: $Bb = [b/x]Bx$ and $\exists x Bx = [a/x]\exists x Bx$

Universal quantifier elimination

Elimination rule $E_\forall$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \forall x \phi & \\ \vdots & & \\ m. & [c/x]\phi & E_\forall, m \end{array}$$

A.k.a. “Universal Instantiation”

Examples

Example 1:

$$\forall x \forall y Rxy \vdash Raa$$

1.	$\forall x \forall y Rxy$	A
2.	$\forall y Ray$	$E_\forall, 1$
3.	Raa	$E_\forall, 2$

Note: $[a/x]\forall y Rxy = \forall y Ray$ and $[a/y]Ray = Raa$

Examples (2)

Example 2:

$$\forall x \forall y Rxy \vdash Rab$$

1.	$\forall x \forall y Rxy$	A
2.	$\forall y Ray$	$E_\forall, 1$
3.	Rab	$E_\forall, 2$

Note $[a/x]\forall y Rxy = \forall y Ray$ and $[b/y]Ray = Rab$

Universal quantifier introduction

Introduction rule $I_\forall$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & [c/x]\phi & \\ \vdots & & \\ n. & \forall x \phi & I_\forall, m \end{array}$$

provided that:

1. *c does not occur in any active assumptions; and*
2. *c does not occur in ϕ*

A.k.a. “Universal Generalisation”

Condition 1

Illustration of condition 1:

1. c does not occur in any active assumptions

$$\begin{array}{lll} 1. & Pa & \mathtt{A} \\ 2.^{\star} & \forall x Px & I_{\forall}, 1 \end{array}$$

Condition 2

Illustration of condition 2:

2 c does not occur in ϕ

1.	$\forall x Rxx$	A
2.	Raa	$E_\forall, 1$
3.*	$\forall x Rax$	$I_\forall, 2$

Examples

$$\forall x(Ax \wedge Bx) \vdash \forall xAx \wedge \forall xBx$$

1.	$\forall x(Ax \wedge Bx)$	A
2.	$Aa \wedge Ba$	$E_\forall, 1$
3.	Aa	$E_\wedge, 2$
4.	Ba	$E_\wedge, 2$
5.	$\forall xAx$	$I_\forall, 3$
6.	$\forall xBx$	$I_\forall, 4$
7.	$\forall xAx \wedge \forall xBx$	$I_\wedge, 5, 6$

Examples(2)

$$\forall x \forall y Rxy \vdash \forall x \forall y (Rxy \wedge Ryx)$$

1.	$\forall x \forall y Rxy$	A
2.	$\forall y Ray$	$E_\forall, 1$
3.	Rab	$E_\forall, 2$
4.	$\forall y Rby$	$E_\forall, 1$
5.	Rba	$E_\forall, 4$
6.	$Rab \wedge Rba$	$I_\wedge, 3, 5$
7.	$\forall y (Ray \wedge Rya)$	$I_\forall, 6$
6.	$\forall x \forall y (Rxy \wedge Ryx)$	$I_\forall, 7$

Examples (3)

$$\neg\exists x A x \vdash \forall x \neg A x$$

1.	$\neg\exists x A x$	A
2.	$A a$	A
3.	$\exists x A x$	$I_\exists, 2$
4.	$\bot$	$E_\neg, 1, 3$
5.	$\neg A a$	$I_\neg$
6.	$\forall x \neg A x$	$I_\forall, 5$

Existential quantifier elimination

Elimination rule $E_\exists$:

$$\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & \exists x \phi & \\
\vdots & & \\
m_2. & [c/x]\phi \to \psi & \\
\vdots & & \\
k. & \psi & E_\exists, m_1, m_2
\end{array}$$

provided that:

1. *c does not occur in any active assumptions; and*
2. *c does not occur in ϕ; and*
3. *c does not occur in ψ*

Condition 1

Illustration of condition 1:

1. c does not occur in any active assumptions

$1.$	$\exists x \exists y Rxy$	$\mathtt{A}$
$2.$	$\exists y Ray \rightarrow Bb$	$\mathtt{A}$
$3.^{\star}$	Bb	$E_{\exists}, 1, 2$

Condition 2

Illustration of condition 2:

2. c does not occur in ϕ

1.	$\forall x \exists y Rxy$	A
2.	$\exists y Ray$	$E_\forall, 1$
3.	Raa	A
4.	$\exists x Rxx$	$I_\exists, 3$
5.	$Raa \to \exists x Rxx$	$I_\to$
$6.^\star$	$\exists x Rxx$	$E_\exists 2, 5$

Condition 3

Illustration of condition 3:

3. c does not occur in ψ

1.	$\exists x Rxx$	A
2.	Raa	A
3.	$\exists y Ray$	$I_\exists, 2$
4.	$Raa \to \exists y Ray$	$I_\to$
5.*	$\exists y Ray$	$E_\exists 1, 4$
6.	$\forall x \exists y Rxy$	$I_\forall 5$

Examples

Example 1:

$$\forall x(Ax \to Bx), \exists x Ax \vdash \exists x Bx$$

1.	$\forall x(Ax \to Bx)$	A
2.	$\exists x Ax$	A
3.	Aa	A
4.	$Aa \to Ba$	$E_\forall, 1$
5.	Ba	$E_\to, 3, 4$
6.	$\exists x Bx$	$I_\exists, 5$
7.	$Aa \to \exists x Bx$	$I_\to$
8.	$\exists x Bx$	$E_\exists, 2, 7$

Examples (2)

Example 2:

$$\exists x \neg Ax \vdash \neg \forall x Ax$$

	1.		$\exists x \neg Ax$		A
	2.	$\neg Aa$	A		
	3.	$\forall x Ax$	A		
	4.	Aa	$E_\forall, 3$		
	5.	$\bot$	$E_\neg, 2, 4$		
	6.	$\neg \forall x Ax$	$I_\rightarrow$		
	7.			$\neg Aa \rightarrow \neg \forall x Ax$	$I_\rightarrow$
	8.			$\neg \forall x Ax$	$E_\exists, 1, 7$

Identity introduction

Introduction rule $I_=$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ k. & c = c & I_= \end{array}$$

Examples

Example 1:

$$\vdash \exists x \exists y\, x = y$$

1.	$c = c$	$I_=$
2.	$\exists y\, c = y$	$I_\exists$
3.	$\exists x \exists y\, x = y$	$I_\exists$

since $c = c$ is $[c/y]c = y$ and $\exists y\, c = y$ is $[c/x]\exists y\, x = y$

Identity elimination

Elimination rule $E_=$:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & c = d & \\
\vdots & & \\
m_2. & [c/x]\phi & \\
\vdots & & \\
k. & [d/x]\phi & E_=, m_1, m_2
\end{array}
$$

Examples

Example 1:

$$a = b, b = c \vdash a = c$$

1.	$a = b$	$\mathtt{A}$
2.	$b = c$	$\mathtt{A}$
3.	$a = c$	$E_=, 1, 2$

Examples (2)

Example 2:

$$Pa, \neg Pb \vdash a \neq b$$

1.	Pa	A
2.	$\neg Pb$	A
3.	$a = b$	A
4.	$\neg Pa$	$E_=, 2, 3$
5.	$\bot$	$E_\neg, 1, 4$
6.	$a \neq b$	$I_\neg$

Examples (3)

Example 3:

$$\forall x \forall y(Rxy \to x = y), a \neq b \vdash \neg Rab$$

1.	$\forall x \forall y(Rxy \to x = y)$	A
2.	$a \neq b$	A
3.	$\forall y(Ray \to a = y)$	$E_\forall, 1$
4.	$Rab \to a = b$	$E_\forall, 3$
5.	Rab	A
6.	$a = b$	$E_\to, 4, 5$
5.	$\bot$	$E_\neg, 2, 6$
6.	$\neg Rab$	$I_\neg$

Homework

Homework

- Read chapter 19 and make exercises 1(c), 2(b), 3(c), 4(a), 5(c
- Read Strawson's 'On Referring'

Soundness and Completeness

Background

Validity and its explications

Semantic validity and syntactic derivability both aim to explain *intuitive validity*.

Observations:

- Syntactic derivability implies intuitive validity:
  we can see that all the steps in a derivation are intuitively correct
- Intuitively validity implies semantic validity:
  if an argument is semantically not valid, then it is not intuitively valid: a counter-example convinces us of that

Validity and its explications (cont.)

Remaining questions:

- Is syntactic derivability not too restricted, i.e., is everything that is intuitively valid also syntactically derivable?
- Is semantic validity not too broad, i.e., is everything that is semantically valid also intuitively valid?

Together, soundness and completeness answer these questions positively: the three notions coincide.

Meta-theorems

The soundness and completeness theorems are so-called ‘meta-theorems’, i.e., not theorems *in* (or *of*) a logical system, but *about* such a system.

Meta-theorems are based on reasoning about properties of a logical system, such as its semantics and its proof theory.

Meta-theorems: semantics

In the semantics of proposition logic and of predicate logic we have already encountered examples of meta-theorems.

Proposition logic:

1. If $\phi \to \psi$ is a contradiction, then ϕ is a tautology and ψ is a contradiction
2. $\{\dagger\}$ is a functionally complete set of connectives

Predicate logic:

1. If $\models \phi$ and $\models \phi \to \psi$, then $\models \psi$
2. $\phi \leftrightarrow \psi \models \chi \leftrightarrow [\phi/\psi]\chi$ (Extensionality)

Meta-theorems: deduction

We can also formulate and prove meta-theorems about proofs:

1. If $\phi \vdash \psi$ and $\psi \vdash \chi$, then $\phi \vdash \chi$
2. If $\neg\phi \vdash \phi$, then $\quad \vdash \phi$
3. If $\Gamma \vdash \psi$, then $\phi_1 \ldots \phi_n \vdash \psi$ for a finite set $\{\phi_1 \ldots \phi_n\} \subseteq \Gamma$
4. Suppose $\phi_1 \ldots \phi_n \vdash \psi$. Then there is a derivation of ψ that starts with the assumptions $\phi_1 \ldots \phi_n$ in which after step n no assumptions occur that are not withdrawn.

The Soundness Theorem

Theorem (Soundness Theorem)

If $\Gamma \vdash \phi$, *then* $\Gamma \models \phi$

Completeness theorem: direct formulation

Theorem (Completeness Theorem (direct))

If $\Gamma \models \phi$, *then* $\Gamma \vdash \phi$

Observations:

- direct proof is do-able for proposition logic
- more difficult for predicate logic:
  - entailment (validity) quantifies over a potentially infinite collection of potentially infinite models

Completeness theorem: indirect formulation

(Henkin) Prove completeness via contraposition:

Theorem (Completeness Theorem (indirect))

If $\Gamma \nvdash \phi$, *then* $\Gamma \not\models \phi$